UNIVERSITY OF SASKATCHEWAN
School of Public Health
SPH.USASK.CA

Primary Prevention in British Columbia: Preliminary Findings on Quantitative Cancer Risk Indicators

T. Nguyen[1], R. Murphy[2], J. Lui[3]

[1] School of Public Health, University of Saskatchewan, [2] Cancer Control Research, BC Cancer, [3] Prevention, Screening and Hereditary Cancer Program, BC Cancer

Background

Modifiable lifestyle factors, environmental exposures and infections contribute to the development of many cancer types[1]

- 4 in 10 cancer cases in Canada can be prevented through healthy living behaviours and supportive policies[2]
- 8,500 cancer cases were preventable in British Columbia in 2015[2]
- 33-37% of incident cancer cases among adults aged 30 years and over in Canada were attributable to preventable risk factors[3]

Exposure	Incident Cases	Exposure	Incident Cases
Tobacco Smoking	17.5%	Alcohol Consumption	1.8%
Physical Inactivity	4.9%	Ultraviolet Radiation	2.3%
Sedentary Behaviour	1.7%	Outdoor Air Pollution ($PM_{2.5}$)	0.9%
Excess Weight	3.1%	Residential Radon	0.9%
Low Vegetable Consumption	0.3%	Hepatitis B Virus	0.1%
Red Meat Consumption	0.6%	HPV	2.0%
Processed Meat Consumption	0.6%	All Exposures (including those not shown)	33.3%

Data Source: ComPARe Study, 2015

- Opportunities to reduce impacts of cancer risk factors exist through implementation of evidence-based policies and programs at the individual, community and societal levels[4]
- Similar to Cancer Care Ontario, BC Cancer intends to develop a report of quantitative indicators on policies and programs addressing cancer risk factors and exposures in B.C.

Objectives

- Conduct background research on potential quantitative cancer prevention indicators, identify data sources, evaluate relevance and quality of indicators, link indicators to public health goals for cancer prevention and propose a reporting framework specific to B.C.

Research Questions:

1. What are the public health programs and policies aimed at cancer risk factor X in Canada and B.C.?
2. What is the prevalence of cancer risk factor X?
3. Are current programs and policies addressing cancer risk factor X effective as evidenced through evaluations?

Methods

An environmental scan and literature review were conducted addressing eight risk factor domains:

- Search strategies were generated using the ECLIPSE framework
- Keywords were combined using Boolean operators in Google Scholar and PubMed
- Environmental scan assessed internal (n = 62) and external (n = 64) gray literature to BC Cancer
- Literature review incorporated title/abstract screening (n = 131) and full-text screening (n = 67) of peer-reviewed articles
- Relevant data was extracted, and key themes collated into a summary report for further analysis

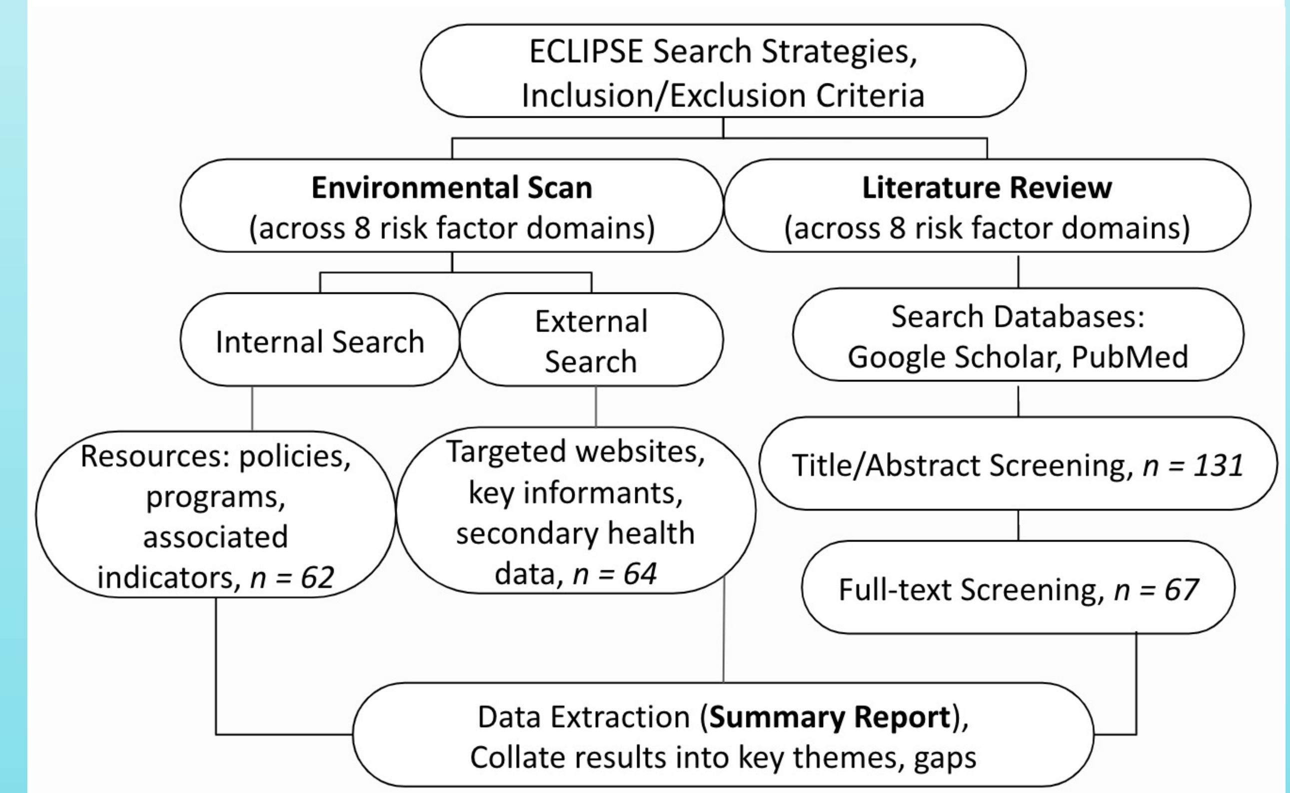
Data Extraction

Relevant data from resources that met inclusion criteria was subdivided into the following categories:

Data Extraction Categories:
1. Prevalence and risk-specific cancer burden
2. Economic costs
3. Targeted policies and programs
4. Indicators of policy/program implementation
5. Evaluation methods and results of policies/programs
6. Limitations and gaps

Methods

Methods Flowchart:



Findings

As recognized by the Canadian Cancer Society and World Cancer Research Fund International, priority cancer risk factor domains identified in this project are:

- Smoking
- Alcohol
- Physical Activity
- Diet
- UV Radiation
- Environmental Exposures
- Occupational Exposures
- Infections

Risk factor-specific results for B.C. showcased in the summary report include indicators identified within the following areas:

- Prevalence data
- Healthy living guidelines
- Harm reduction policies
- Behavioural interventions
- Public health programming at the community, municipal, provincial and national levels

Recommendations

- Operationalization of the built environment and community-based programs and policies
- Stratification of cancer risk indicators (geographical residence, household income, education, neighbourhood, etc.) as social determinants of health can highlight disparities impacting indicators

Conclusion

- Interventions targeting modifiable cancer risk factors may contribute to the prevention of 10,600 to 39,700 cases in Canada by 2042[2]
- Findings from the environmental scan and literature review will require further synthesis as priority indicators are identified by BC Cancer
- **As health care provision is decentralized in Canada, this project highlights the importance of prioritizing the development and implementation of cancer prevention strategies specific to B.C.**

Acknowledgements

My sincere gratitude to Dr. Rachel Murphy and Javis Lui at BC Cancer for their guidance and expertise throughout the practicum.
This work was also accomplished with the support of key informants from BC Cancer Health Promotion, the BC Centre for Disease Control and the BC Cancer Registry.

Key References

1. Pader, J., Ruan, Y., Poirier, A. E., Asakawa, K., Lu, C., Memon, S., ... & Brenner, D. R. (2021). Estimates of future cancer mortality attributable to modifiable risk factors in Canada. *Canadian Journal of Public Health*, *112*(6), 1069-1082.
2. Poirier, A. E., Ruan, Y., Volesky, K. D., King, W. D., O'Sullivan, D. E., Gogna, P., Walter, S. D., Villeneuve, P. J., Friedenreich, C. M., Brenner, D. R., & ComPARe Study Team (2019). The current and future burden of cancer attributable to modifiable risk factors in Canada: Summary of results. *Preventive medicine*, *122*, 140–147.
3. ComPARe. (2015). *Canadian Population Attributable Risk of Cancer study on preventable cancers in Canada*. Retrieved July 27, 2022, from https://prevent.cancer.ca/
4. Ontario Health (Cancer Care Ontario). Prevention System Quality Index 2020. Toronto: Queen's Printer for Ontario; 2020.